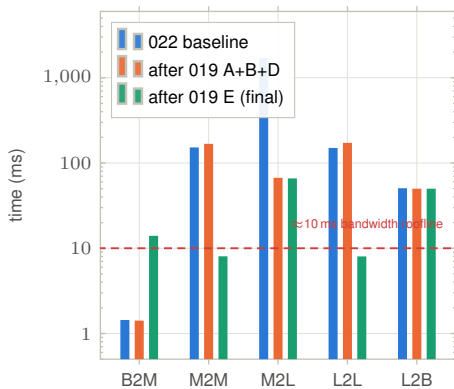
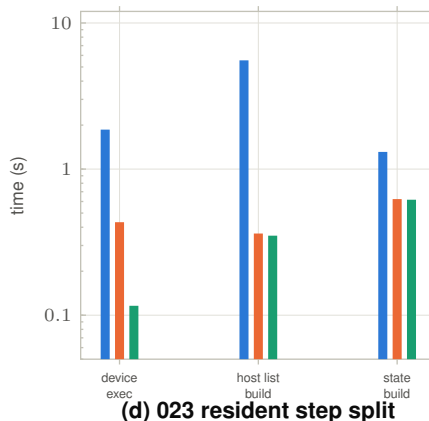


Fig. 3 — The 019 tuning gains stage by stage, what is left between M2L and the roofline, and what the 023 step costs

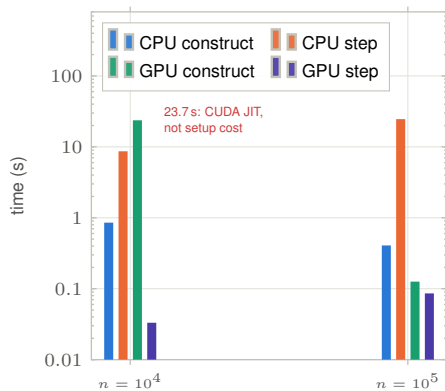
(a) device stage time



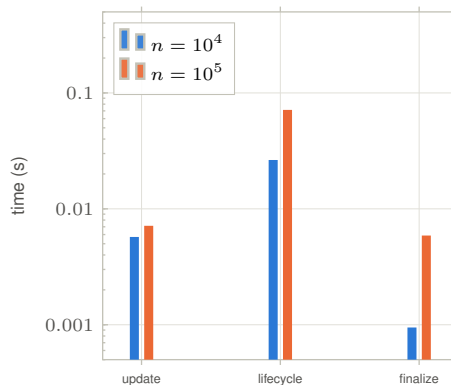
(b) 019 execution and setup costs



(c) 023 one-shot setup vs recurring step



(d) 023 resident step split



Machine / regime: m13h-1-1, NVIDIA H200, CUDA 12.8, Julia 1.11.7. Panels (a,b): $n = 10^5$, $\ell = 4$, $P = 4$, ConstantPAnalyticStencil(4,1e-4), concat chunk 2^{17} . Panels (c,d): task 023 production-integration radix path, $P = 4$, $\ell = 4$, Julia threads=8. **Data:** operator_performance_tuning/cuda_022_baseline.csv, m13h-1-1/cuda_019_phase{ABD,E}_throughput.csv, production_integraton/benchmark_023_m13h-1-1_20260714-233159.csv (tables fig03a_*, fig03b_*, fig03d_* for panel (c), fig03c_* for panel (d)). **Reads as:** M2L fell 1698 $\rightarrow$ 66 ms ($\approx 25\times$) and M2M/L2L $\approx 150 \rightarrow 8$ ms ($\approx 19\times$), taking the whole device lifecycle 1.863 $\rightarrow$ 0.116 s ($\approx 16\times$). M2L is now $\approx 6\times$ above the memory-bandwidth roofline, down from $\approx 100\times$. The largest remaining device stage is L2B at 50 ms, which the tuning did not touch — and which is not a far-field operator at all: L2B evaluates the *nearfield* direct pairs as well as the local expansions, so most of that bar is $O(N^2)$ -within-the-complement work that no operator strategy can remove. In the pre-023 reconstruct-every-call workflow, the 0.35 s host list build and 0.62 s state build would have dominated the 0.116 s device evaluation. Panel (d) shows the actual 023 recurring step: device-side update replaces those builds, so neither setup bar belongs in the recurring loop. Panel (c) prices the remaining one-shot cost against the step it enables: on the GPU the representative cache construction is 0.126 s ($n = 10^5$) against a 0.086 s step, so device residency pays for itself within about two steps, while on the CPU an 0.41 s construction sits in front of a 24.6 s step that is not worth taking at all (Fig. 8). **Caveats:** the 022 “baseline” stage times are *single-shot* while both post-tuning columns are min-of-3, so the $\approx 25\times$ M2L figure mixes timing methods. The post-E B2M bar (14 ms vs 1.4 ms before) is measurement noise, not a regression — task 019 recorded B2M jitter of 8–26 ms across sweep rows, and the other two chunk widths in the same phase-E sweep read 1.4 ms. Panel (c)’s GPU construction bar at $n = 10^4$ (23.7 s) is *not* a setup cost: that case runs first in the 023 script, so it absorbs one-time CUDA kernel compilation — the same host reads 0.126 s at $n = 10^5$ and m13h-2-1 reads 24.7 s then 0.030 s for the same pair. Read the $n = 10^5$ bar as the representative construction cost. **Scope:** panel (b)’s device exec and the step bars of panels (c,d) are complete `fmm!` evaluations — far field *and* nearfield, the latter inside L2B — so they are end-to-end influence, not a far-field-only subset; task 024 validates the same pipeline against an all-bodies `direct!` reference. The chunk width itself changes the lifecycle minimum by at most $\approx 6\%$; 2^{17} was kept for least scratch, not because it was fastest.